

Abstract

Our project focuses on predicting customer status for a South Carolina company using Probabilistic Graphical Model (PGM) while addressing challenges associated with imbalanced data through different sampling methods. The project's objective is to enhance the accuracy and reliability of predictions, especially in scenarios where imbalances in the dataset could impact the model's performance. We will introduce Bayesian networks, a type of PGM and the implementation of k-fold validation. Explorations into evaluation metrics provide a nuanced understanding of the model's performance. The final stages of the project involve creating graphical representations of key metrics, exploring UnderSampling and OverSampling techniques. The research contributes valuable insights into predicting customer status, particularly in the context of imbalanced data, with the outcomes and methodologies poised for discussion and application in practical settings.

Background

A **Bayesian network** (Fig. 1) is a probabilistic graphical model that represents a set of variables and their conditional independencies using directed acyclic graphs. Bayesian networks are widely used in fields such as machine learning, artificial intelligence, and decision analysis.

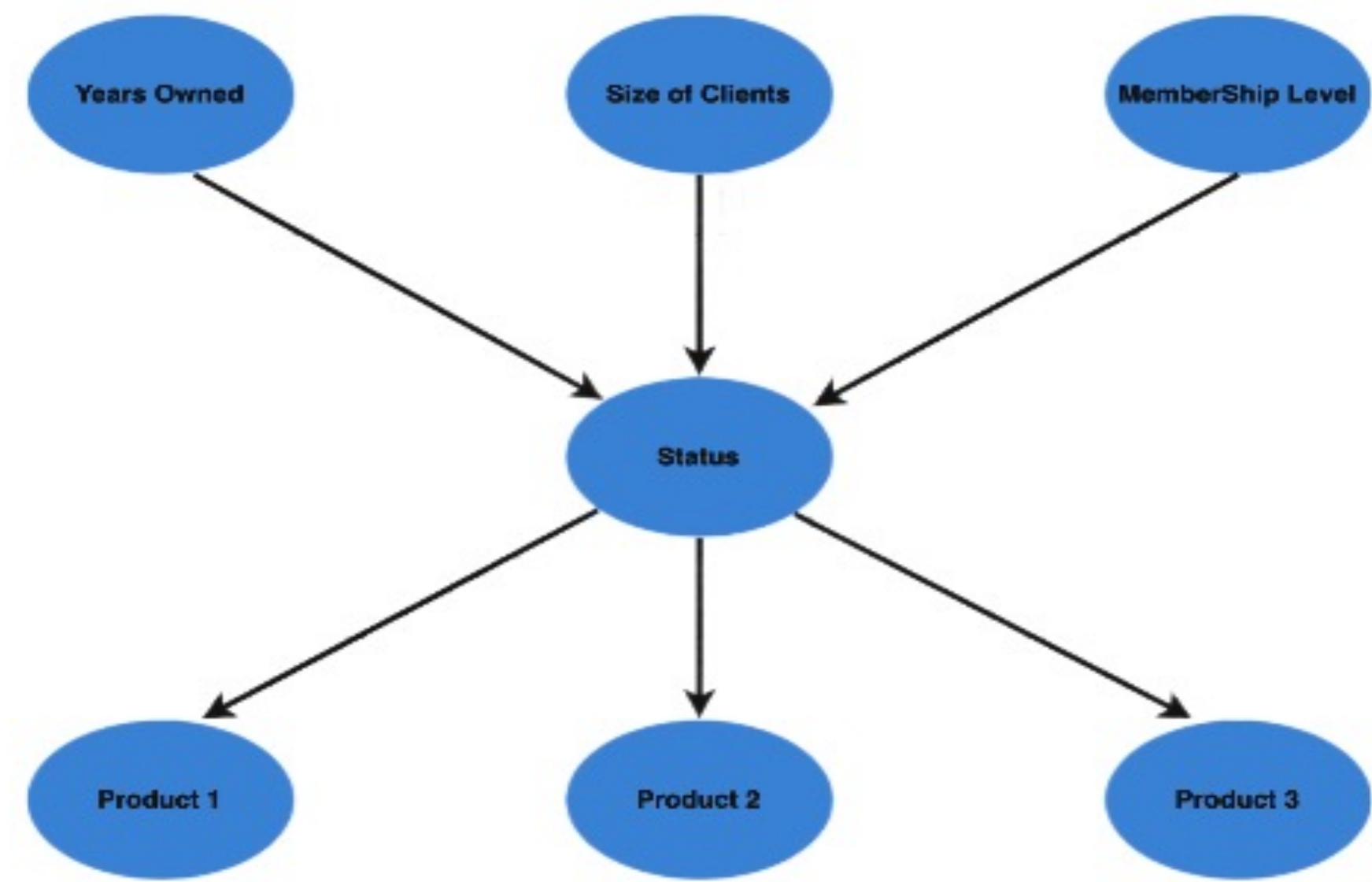


Fig.(1)

A **Confusion matrix** (Fig. 2) represents the prediction summary in matrix form. It shows how many prediction are correct and incorrect per class.

Evaluation metrics (Fig. 3) are quantitative measures used to assess the performance and effectiveness of a statistical or machine learning model. These metrics provide insights into how well the model is performing and help in comparing different models or algorithms.

		Actual Values	
		Positive (1)	Negative (0)
Predicted Values	Positive (1)	TP	FP
	Negative (0)	FN	TN

Fig.(2)

$$\text{Accuracy} = \text{Correct Prediction Rate} = \frac{TP+TN}{TP+TN+FP+FN}$$

$$\text{Sensitivity} = \text{True Positive Rate} = \frac{TP}{TP+FN}$$

$$\text{Specificity} = \text{True Negative Rate} = \frac{TN}{TN+FP}$$

$$\text{Precision} = \text{Positive Predictive Value} = \frac{TP}{TP+FP}$$

Fig.(3)

Computations

The calculations were performed on MacBook Air M2 (8 GB memory, 8 core CPU) using python as programming language. Packages like Matplotlib, NetworkX, NumPy, Pandas, Pgmpy, and SciKit-Learn were used.

Methodology

1) The datasets were preprocessed using Python to ensure data quality and consistency.

Imbalanced Dataset

- Used the variables in the Bayesian network to fit the imbalanced dataset and validated the model:
 - Accuracy: 96.414%
 - Standard Deviation: 0.809%
- Customer status for the year 2023 was predicted using the 2022 dataset by incrementing the purchase year by 1.
- The model exhibited poor performance in predicting true negatives, attributed to missing customer data in the 2023 dataset and the high imbalance within the data (86% majority class). See figure 4.
- The missing customer statuses were assumed to be "Not Active," and the model was re-evaluated. See figure 5.

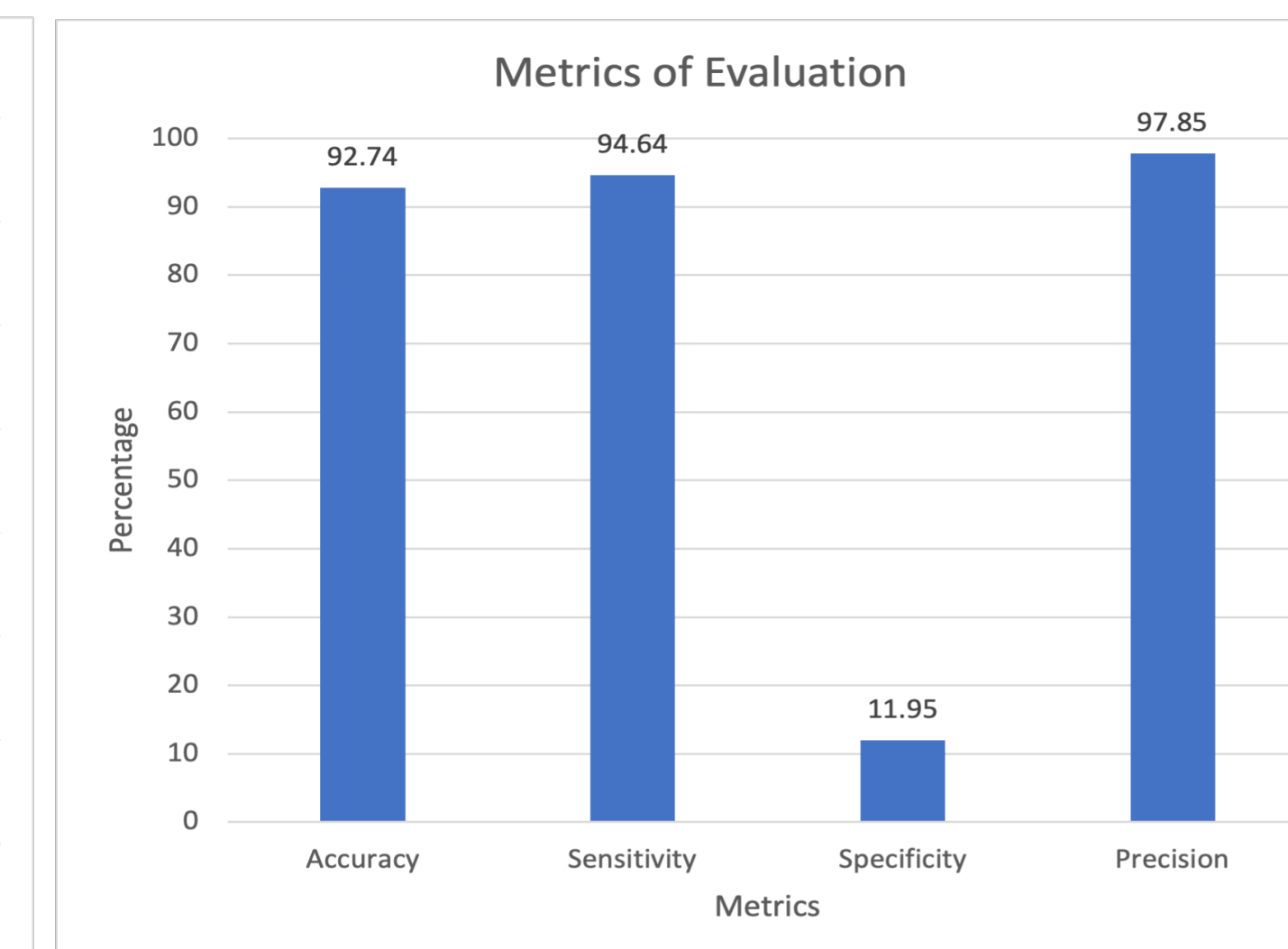
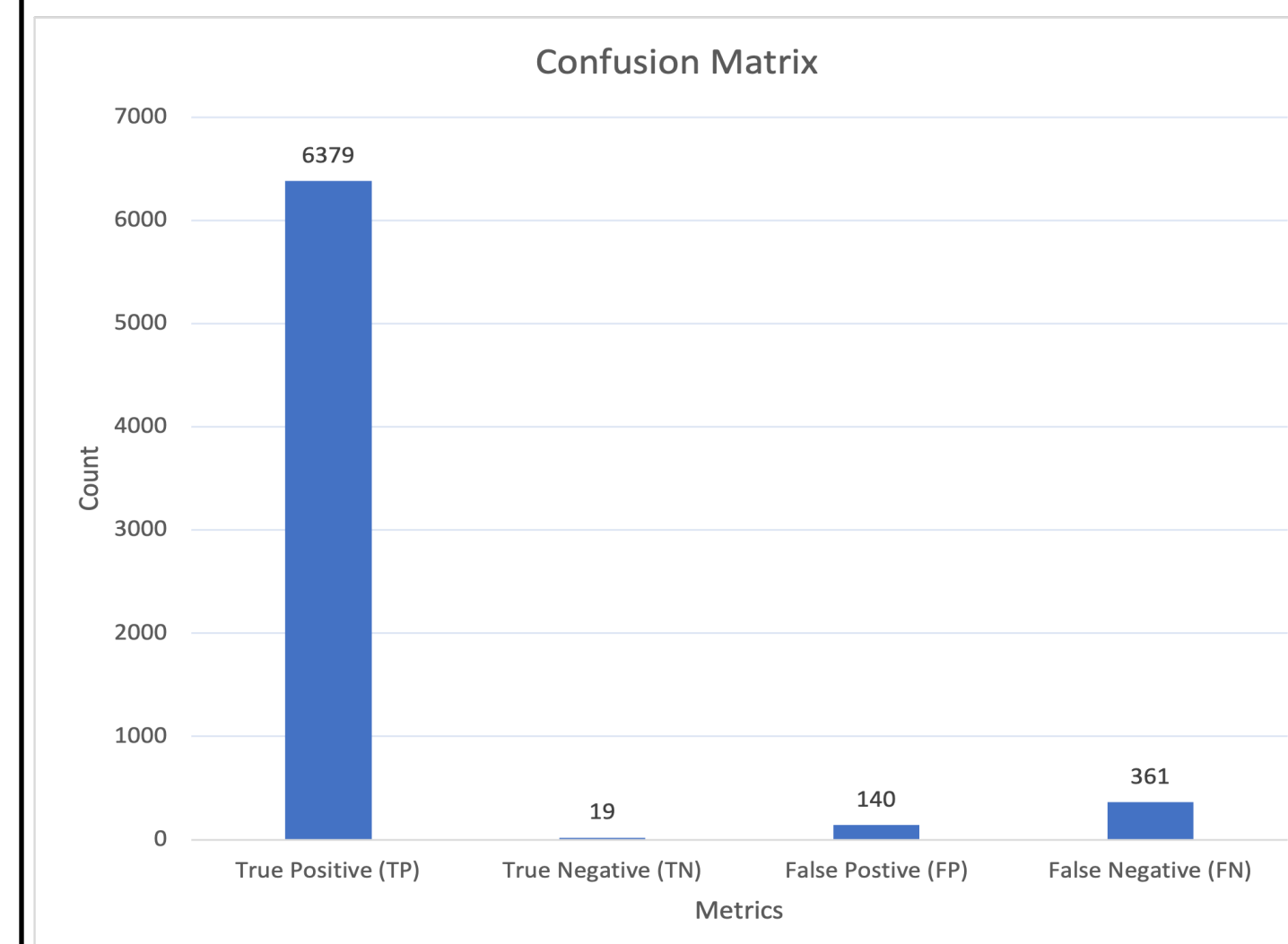


Fig.(4)

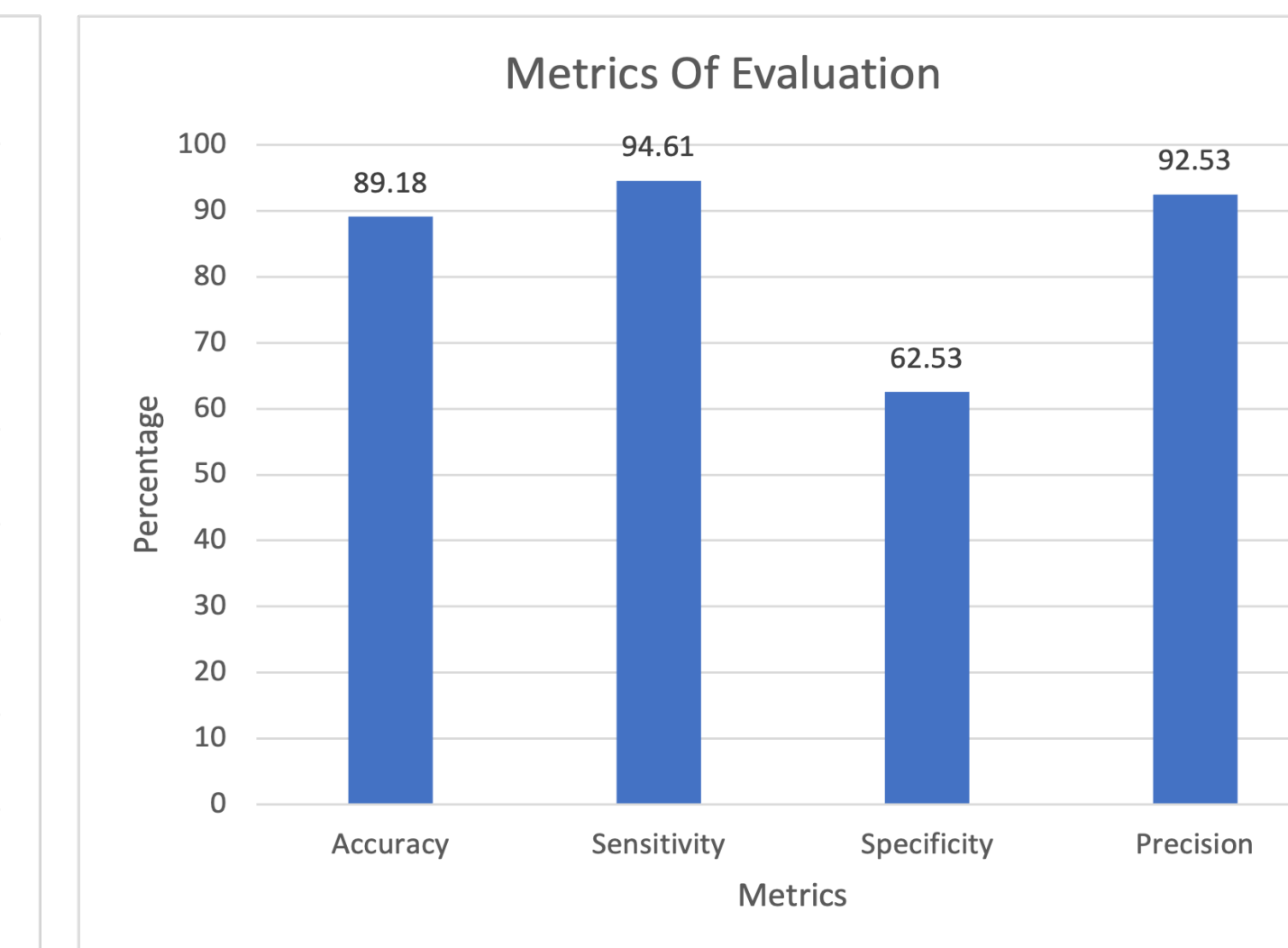
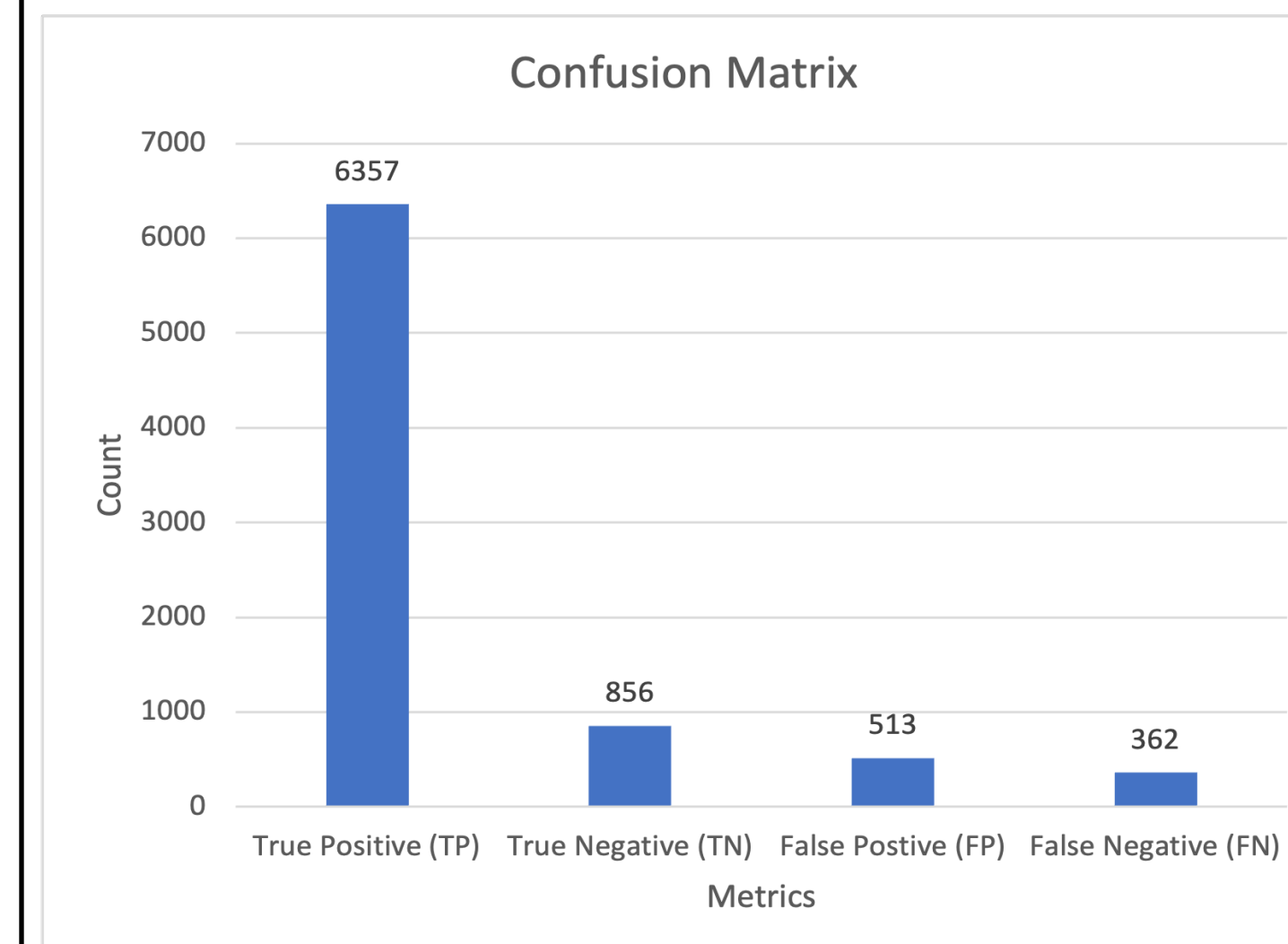
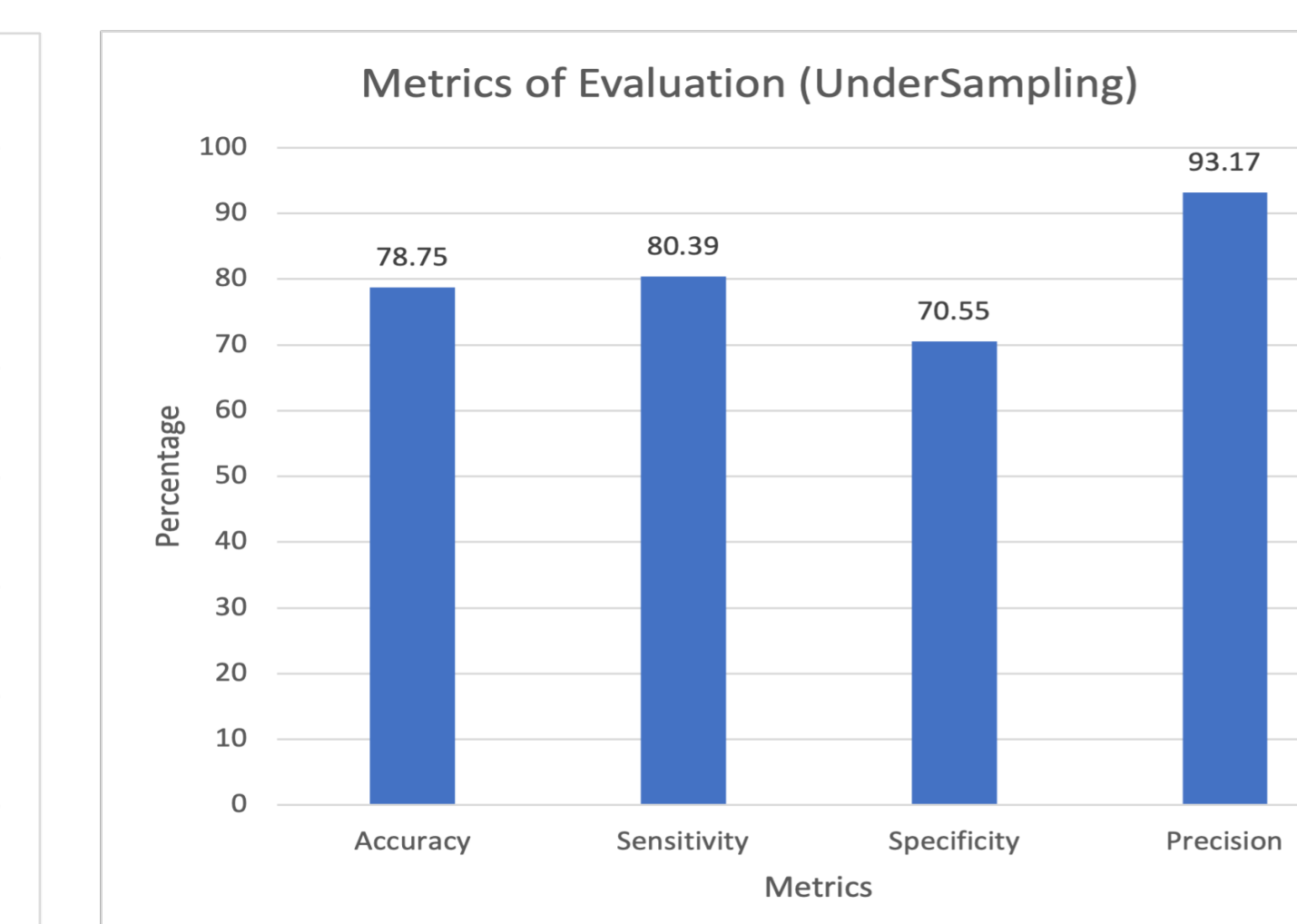
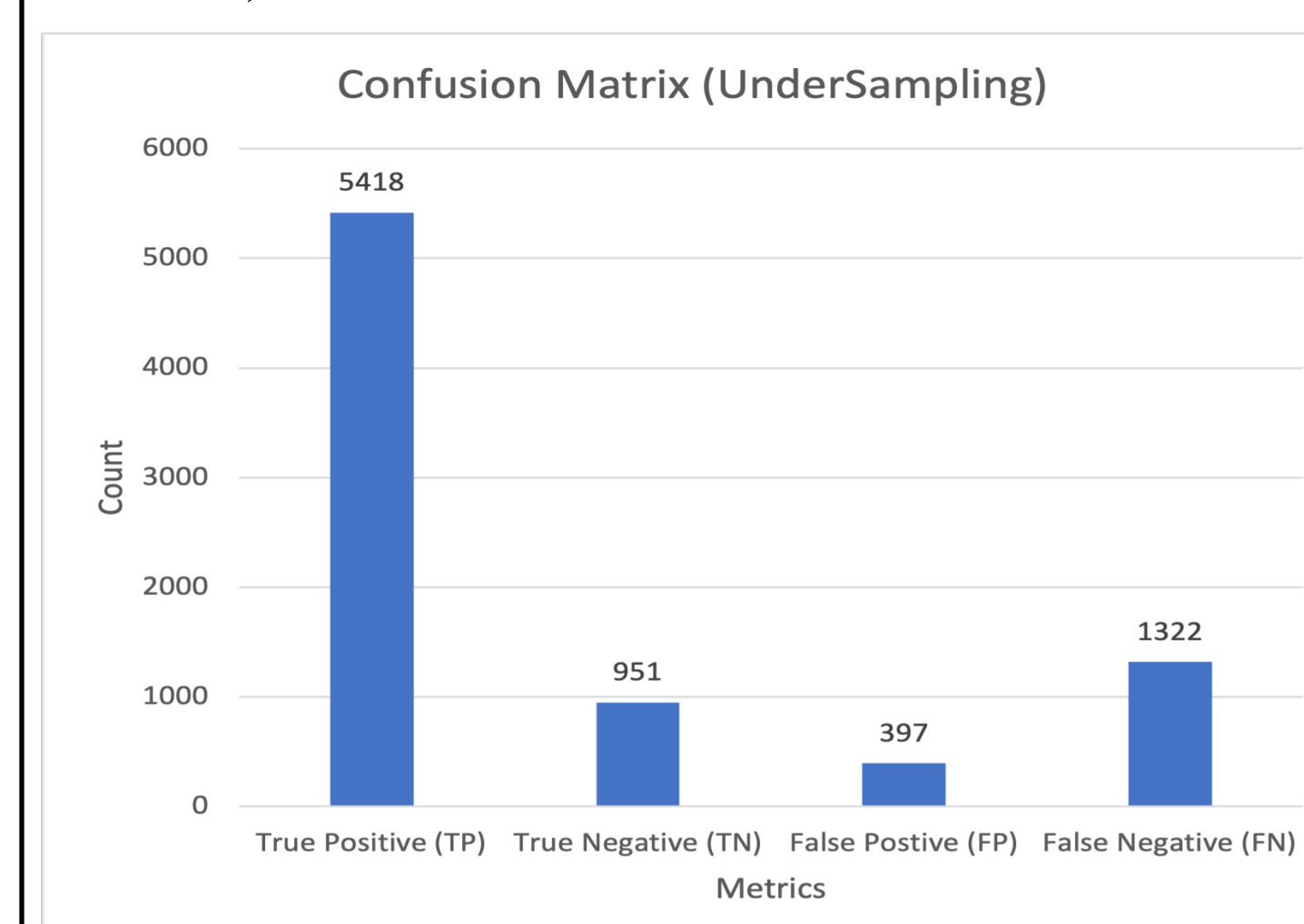


Fig.(5)

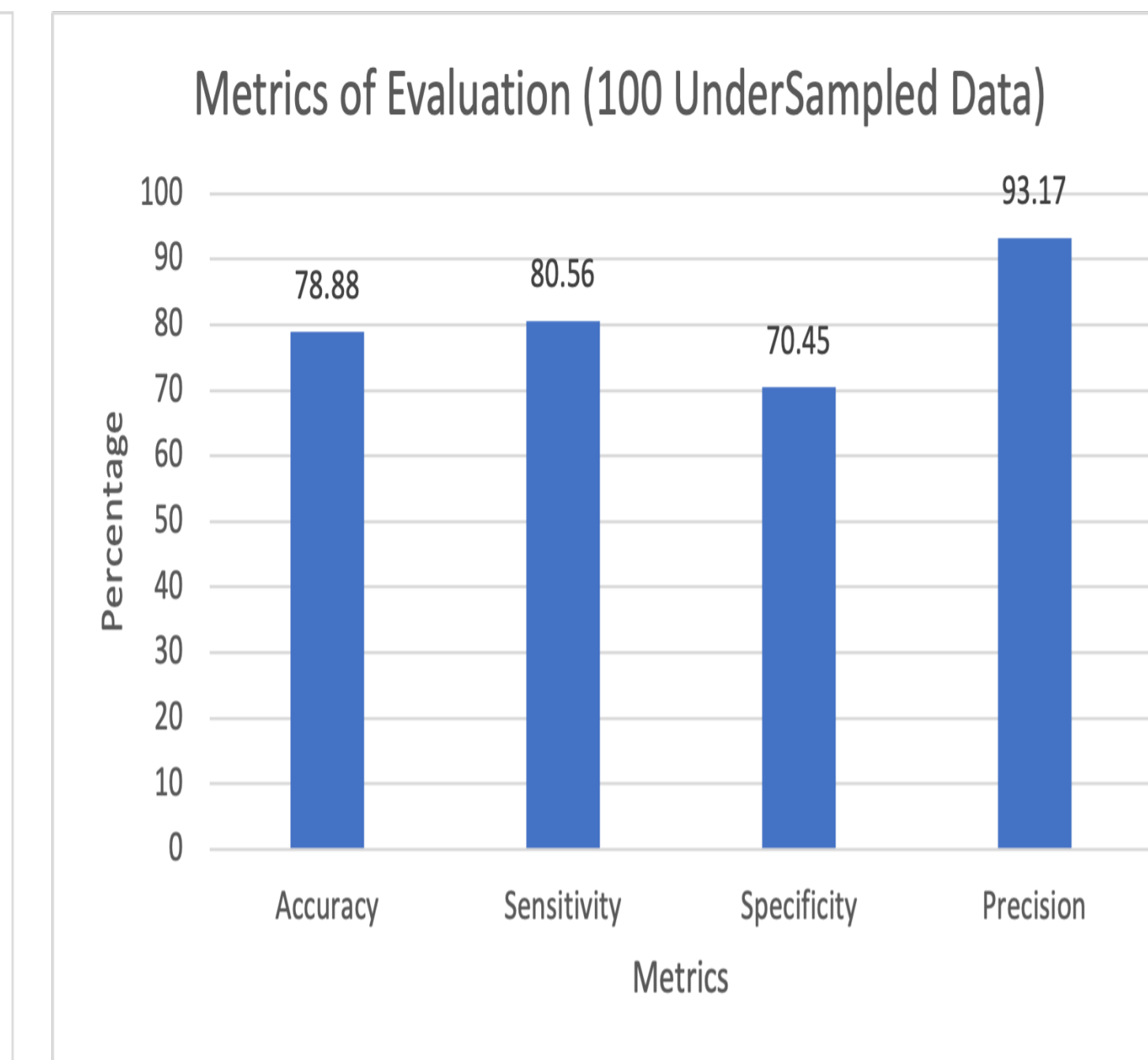
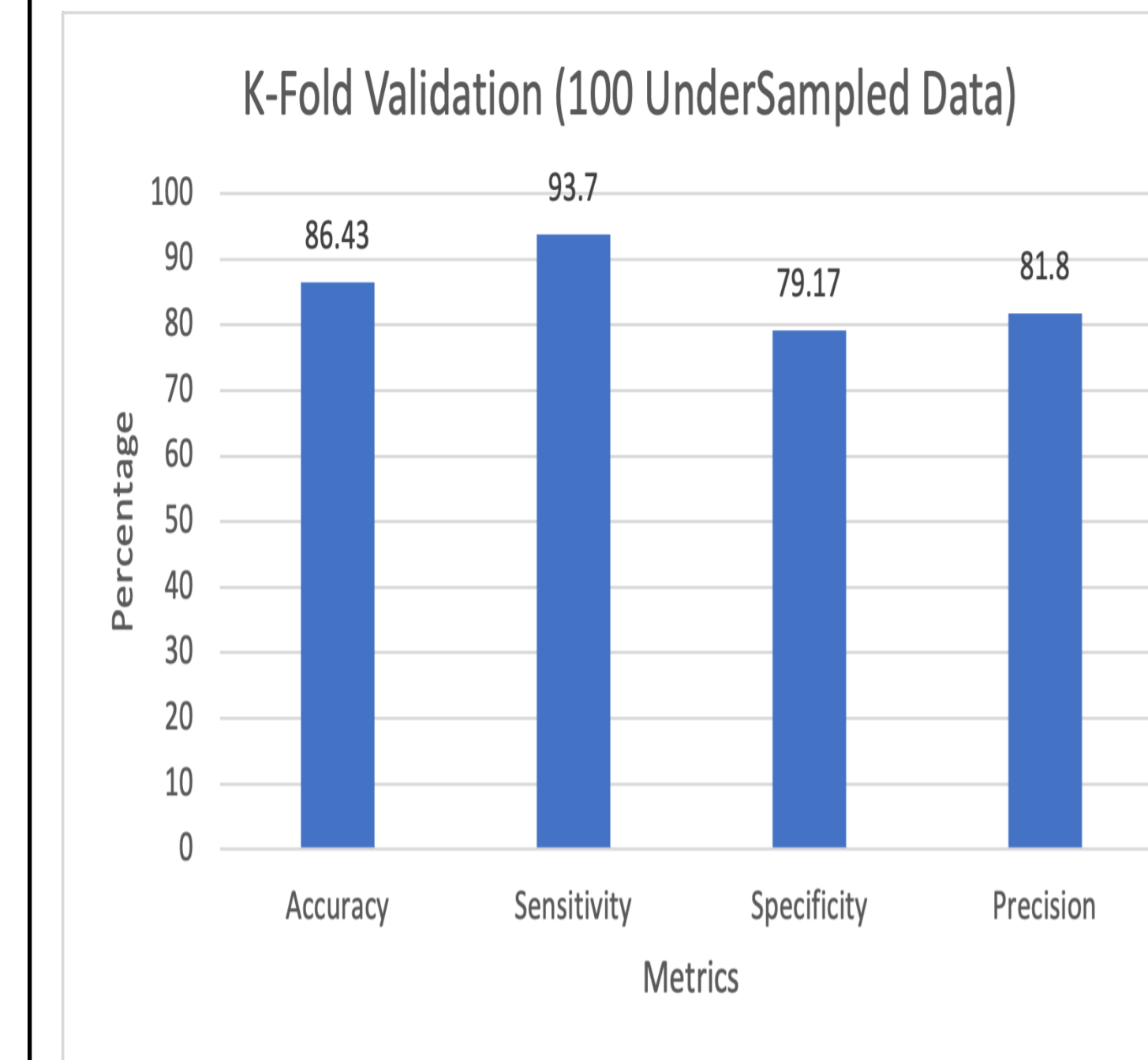
Under Sampled Dataset

Keep the data from the minority class and decrease the size of the majority class to the number of the minority class. Validated the model:

- Accuracy: 86.14%
- Standard Deviation: 3.35%



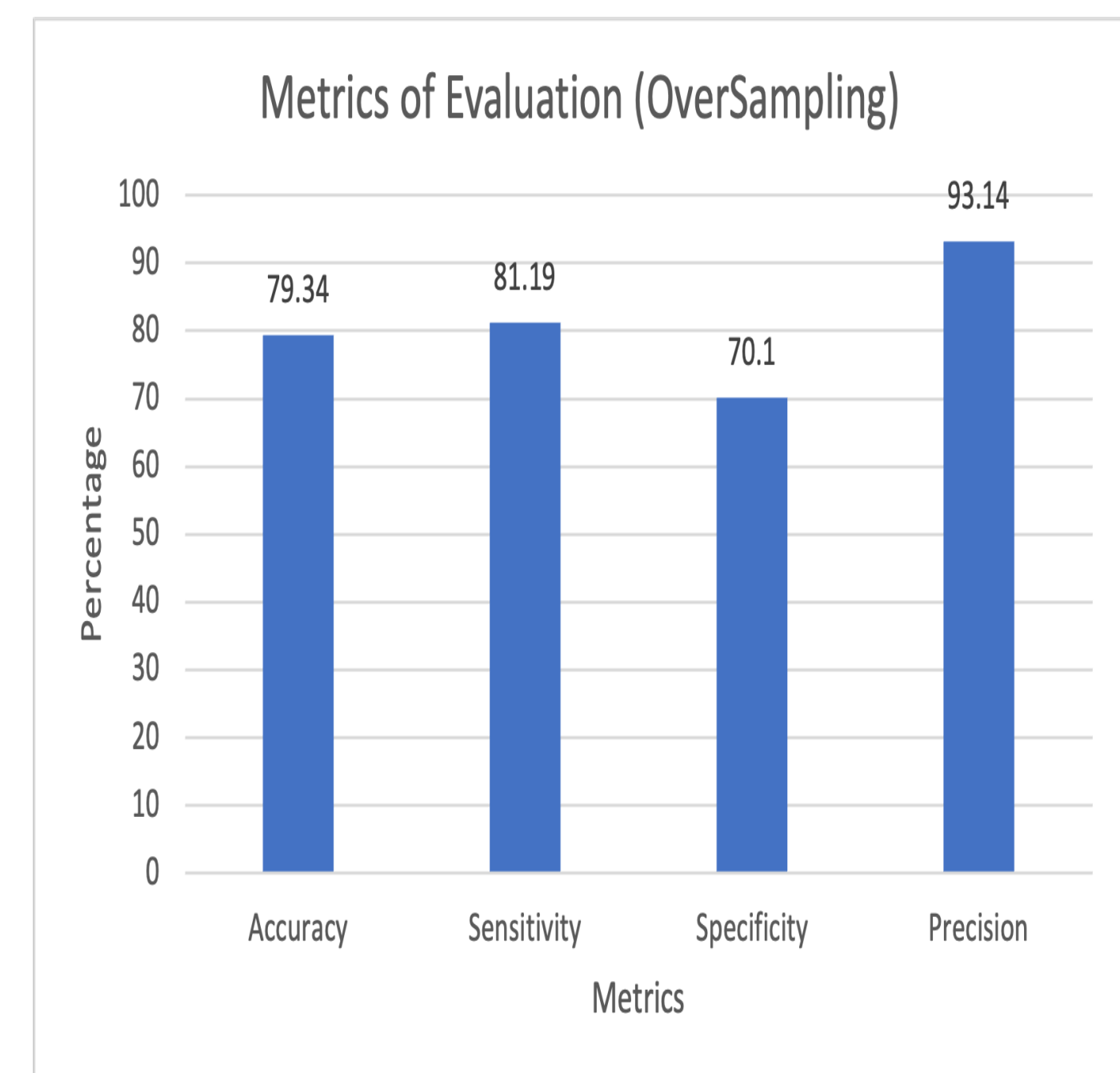
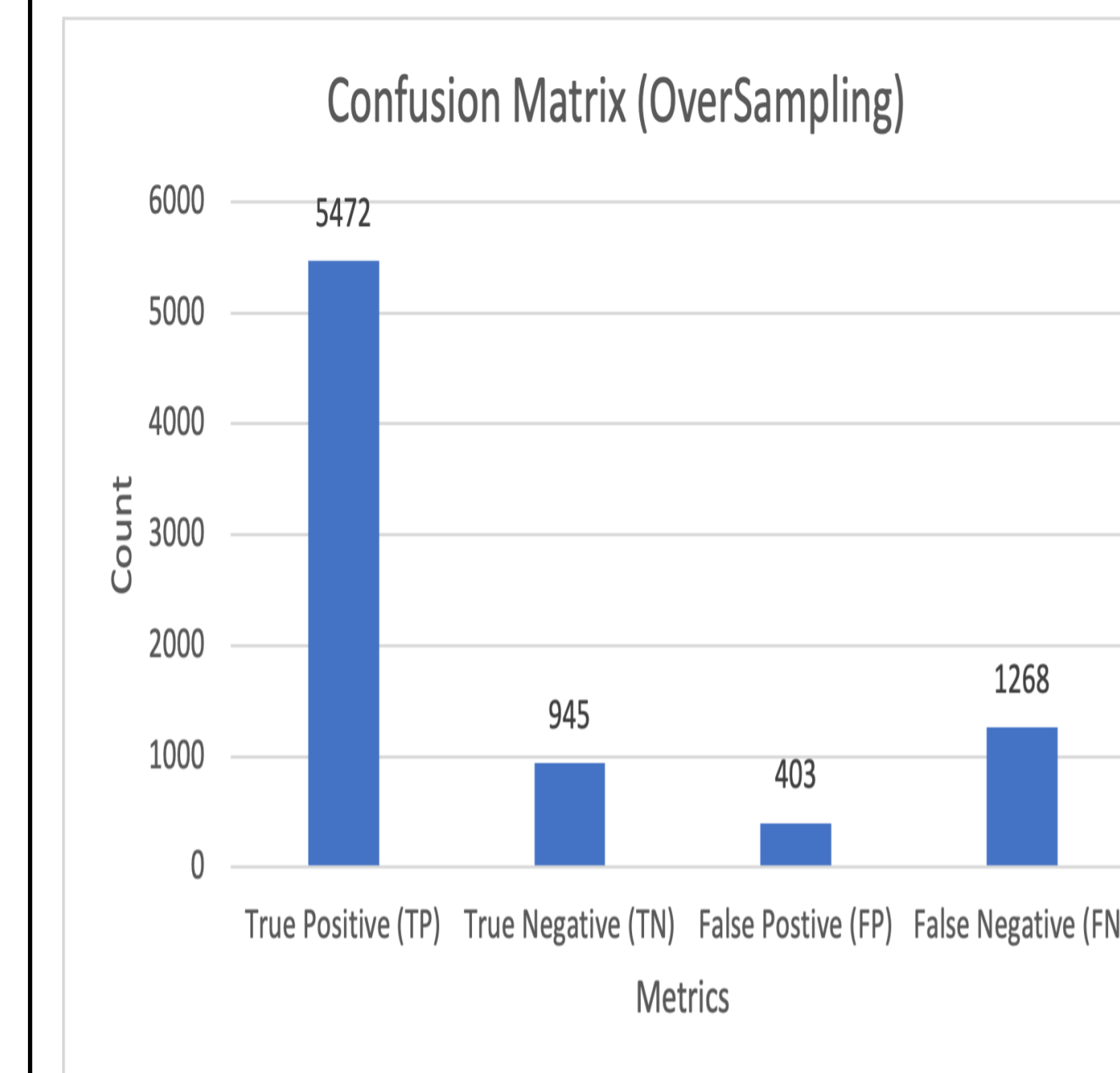
The UnderSampling method's performance was assessed through predictions on 100 datasets with randomization for reproducibility and consistency.



Over Sampled Dataset

Keep the data from the majority class and increase the size of the minority class by duplicating the values. Validated the model:

- Accuracy: 87.05%
- Standard Deviation: 1.08%



Conclusion

Our project highlights the effectiveness of Bayesian Networks in predicting customer status for a South Carolina company despite challenges posed by imbalanced data. While initial validation showed promise, issues in predicting true negatives prompted the adoption of UnderSampling and OverSampling techniques. These methods led to increased prediction accuracy and improved performance metrics. Our evaluation of the UnderSampling method across 100 datasets underscores its reliability and reproducibility, offering valuable insights for practical applications in customer relationship management.

References

- Brownlee, J. (n.d.). Undersampling Algorithms for Imbalanced Classification. Retrieved from <https://machinelearningmastery.com/undersampling-algorithms-for-imbalanced-classification/>
- Scutari M, Denis J (2021). Bayesian Networks with Examples in R, 2nd edition. Chapman and Hall, Boca Raton.

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